

5 (**short**) A two-dimensional jet of water of density  $\rho$ , thickness  $d_1$  and velocity  $V_1$  is deflected by a two-dimensional obstacle forming two diverging jets of uniform velocity  $V_2$ , as shown in Fig. 2. The pressure is the atmospheric pressure throughout and friction and gravity may be neglected throughout.

- (a) Find the velocity  $V_2$  and thickness  $d_2$  of the diverging jets in terms of  $V_1$  and  $d_1$ . [5]
- (b) Find the force that the fluid exerts on the obstacle in terms of  $\rho$ ,  $V_1$  and  $d_1$ . [5]

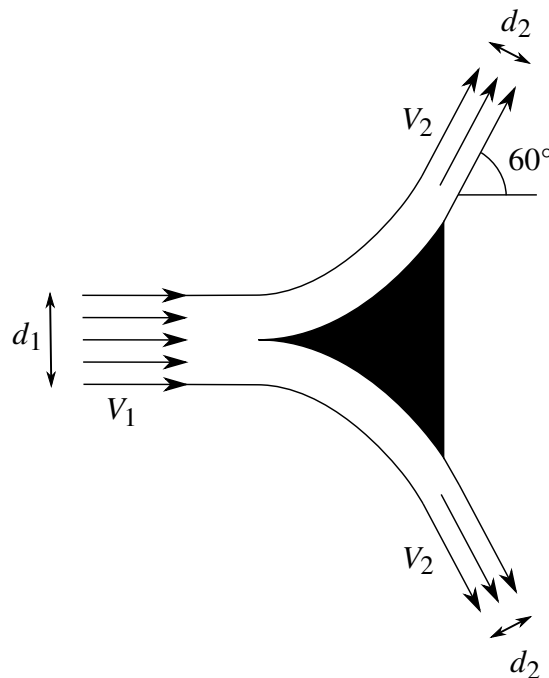


Fig. 2